WELDING QUALITY CONTROL PROCEDURE – STRUCTURAL STEEL

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1. Applicable Standards

This procedure is based on AWS D1.1 (Structural Welding Code – Steel) and ASME Section IX. All welding activities shall comply with the approved Welding Procedure Specification (WPS).

2. Pre-Weld Requirements

- Base material must be free from oil, grease, rust, paint, moisture, and any contaminants.
- For carbon steel thickness greater than 25 mm, minimum preheat temperature is 75°C.
- Welder must possess a valid qualification certificate for the process and position being used.
- Joint fit-up must be inspected and accepted before welding begins.

3. Acceptance Criteria

Visual Inspection Limits:

- Porosity: maximum 2% of the weld surface area in any 25 mm length.
- Undercut: maximum depth 0.5 mm; continuous undercut not permitted.
- Cracks of any size or type are not acceptable.
- Incomplete fusion or incomplete penetration is rejectable.
- Excess reinforcement: maximum 3 mm for plate thickness up to 25 mm.

4. Non-Destructive Testing (NDT) Requirements

- Visual Inspection (VT): 100% of all welds.
- Ultrasonic Testing (UT): minimum 20% of critical butt joints.
- Magnetic Particle Testing (MT): on request or for fillet welds in high-stress areas.
- Radiographic Testing (RT): as specified in the project quality plan.

5. Documentation and Traceability

Every weld must be recorded with the following information: Welder ID, Date, WPS number, Joint number, Welding process, Consumable batch number, and final inspection result (Accept / Reject). Records shall be retained for a minimum of 10 years.

6. Repair Procedure

Defective welds shall be removed by grinding or gouging. The excavated area must be inspected by MT or PT before re-welding. Repair welding shall be performed using the same or a higher qualified WPS. A maximum of two repair attempts is permitted on the same location.

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